

Adaptive Multi-Expert Graph Transformer for Interpretable EEG-Based Diagnostics

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Abstract

Electroencephalographic (EEG) abnormalities arise from dynamic changes in neural synchrony across spatial and temporal scales, yet many computational approaches reduce these dynamics to static features. We present a Spatial Multi-Expert Graph Transformer that models each EEG recording as a sequence of dynamic functional connectivity graphs. Time-resolved connectivity is estimated using the weighted Phase Lag Index (wPLI), and hierarchical graph encoding aggregates information from electrode to regional and global levels. A multi-expert transformer architecture enables subtype-aware reasoning, with a gating mechanism adaptively fusing expert outputs for global abnormality prediction. Experiments on the TUAB dataset show competitive abnormal EEG detection performance and demonstrate the potential of dynamic graph modeling with adaptive expert fusion for interpretable, subtype-aware spatial-temporal analysis.

Keywords: EEG analysis, graph transformers, multi-expert systems, neuroinformatics, interpretable AI, healthcare analytics

1. Introduction

Electroencephalography (EEG) [1] is a widely used non-invasive technique for assessing brain activity in clinical and research settings. EEG plays a critical role in detecting neurological abnormalities such as epileptiform discharges [2], focal slowing, diffuse slowing, and hemispheric asymmetries. EEG interpretation relies on identifying spatially localized and temporally evolving patterns, rather than treating recordings as static signals. However, many automated EEG analysis methods still assign a single global label to an entire recording, limiting their ability to reflect clinically meaningful dynamics.

Recent deep learning approaches have improved EEG abnormality detection using temporal-spectral modeling, transformer-based architectures, and graph-based representations. CNN-LSTM hybrids and spectral networks such as FFCL [3] and SPaRCNet [4] capture short-range temporal dynamics, while transformer-based models model long-range dependencies, including S3T [5], BIOT [6], and LaBraM [7]. Graph-based methods further encode spatial structure through correlation-aware and local-global connectivity modeling, such as Corr-DCRNN [8], LGGNet [9], and XAIguiFormer [10]. Despite these advances, most existing methods rely on static connectivity representations and single-expert reasoning, limiting their ability to capture dynamic and subtype-specific EEG abnormalities.

To address these limitations, this work introduces a Spatial Multi-Expert Graph Transformer that models EEG recordings as sequences of dynamic functional connectivity graphs. Functional connectivity is estimated using the wPLI [11], enabling time-resolved graph representations of evolving neural synchrony. A hierarchical graph encoder aggregates information from electrode-level features to regional and global representations, while a multi-expert transformer enables subtype-aware reasoning through adaptive expert gating. This design is intended to support robustness, interpretability, and alignment with clinical EEG practice.

2. Framework Overview

The proposed framework transforms EEG recordings into sequences of dynamic functional connectivity graphs for subtype-aware abnormality detection through hierarchical graph encoding and a multi-expert temporal transformer (Figure 2). The model employs a three-tier spatial hierarchy, progressing from electrode-level encoding to regional aggregation and global fusion for classification. Figure 3 illustrates connectivity aggregation from individual electrodes to regional and global representations.

EEG recordings are preprocessed using a 1–40 Hz bandpass filter, artifact attenuation, and resampling to 250 Hz. Each recording is divided into overlapping temporal windows, with each window converted into a functional connectivity graph using the wPLI. Nodes represent electrodes and weighted edges encode pairwise interactions, producing a time-

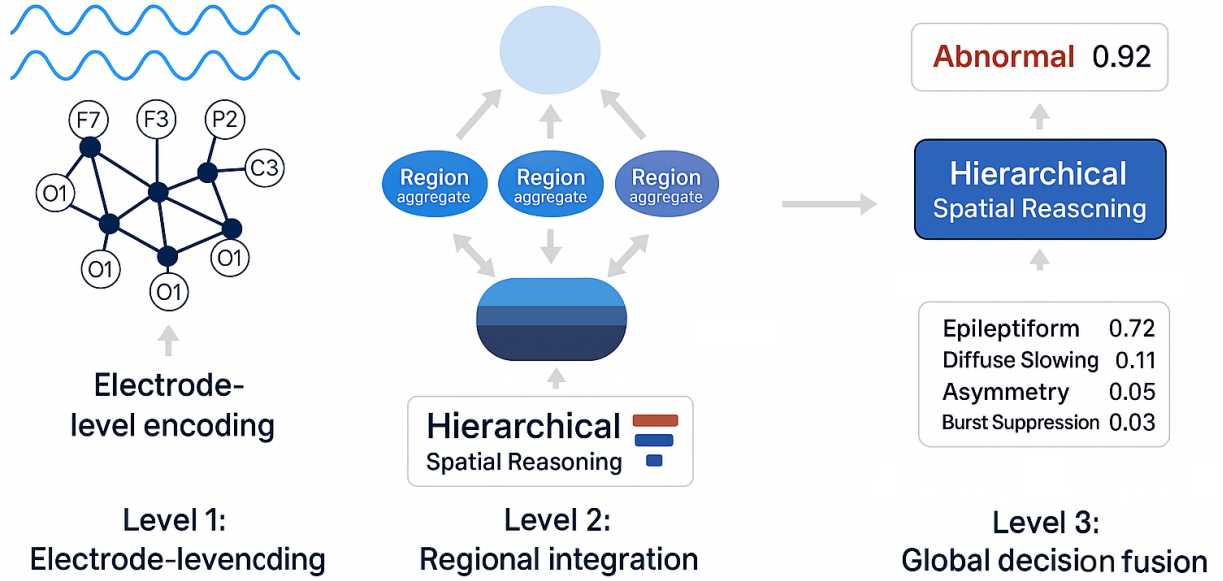


Figure 1: Hierarchical spatial encoding and multi-expert temporal reasoning framework.

ordered sequence of dynamic brain graphs. A hierarchical graph encoder processes each graph at three resolutions: electrode-level graph convolutions extract local features, regional pooling models meso-scale interactions, and global aggregation summarizes brain-wide connectivity patterns.

The encoded graph sequence is processed by a temporal transformer modeling long-range dependencies. A multi-expert module with specialized heads for distinct abnormal subtypes receives transformer outputs, with a learned gating network assigning adaptive expert weights based on input representations. Fused expert outputs generate subtype-specific probabilities and global abnormality predictions, enabling interpretable reasoning with subtype-specific contributions.

Algorithm 1 summarizes the end-to-end pipeline, detailing dynamic graph construction, hierarchical encoding, temporal modeling, and gated multi-expert decision fusion.

2.1 Implementation Details

Dataset and splits. All experiments were conducted on the Temple University Hospital (TUH) EEG corpus [12], a large-scale clinically curated dataset that has been widely adopted for abnormal EEG detection and seizure analysis [3, 4, 9]. Data were split into training (80%) and test (20%) sets with stratification by class. Splits were performed at the recording level; patient-level metadata were not available.

Windowing and graph construction. Each EEG recording was resampled to 250 Hz and segmented into $S = 6$ non-overlapping temporal windows of equal length. For each window, a functional connectivity graph was constructed using wPLI in the 4–30 Hz band. When wPLI computation was unstable, Pearson correlation was used as a fallback. Graphs were fully connected with self-loops removed.

Node features. Each node (electrode) was represented using three normalized channel-wise statistics computed per window: mean amplitude, standard deviation, and peak-to-peak range. Hjorth parameters and entropy were used only for preliminary analysis and not as model inputs.

Model configuration. The hierarchical graph encoder used three GCN layers with hidden dimension 128, interleaved with self-attention graph pooling (SAGPool). The temporal module consisted of a 2-layer transformer encoder with 4 attention heads. The full model employed $K = 7$ subtype-specific experts with a learned gating mechanism.

Training. Models were trained using AdamW (learning rate 5×10^{-4} , weight decay 10^{-4}) for 15 epochs. The objective combined global classification loss, expert loss, subtype prediction loss, and hierarchical consistency regu-

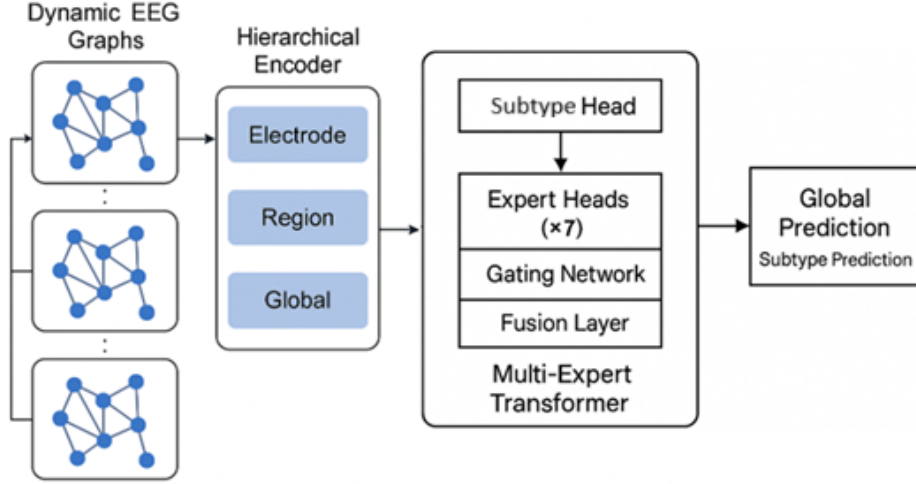


Figure 2: Pipeline of dynamic EEG graph construction, hierarchical encoding, and the multi-expert transformer for subtype-aware and global EEG abnormality detection.

larization.

3. Results and Analysis

This section evaluates the proposed Spatial Multi-Expert Graph Transformer, including baseline performance, hyper-parameter sensitivity, optimization effects, and final model behavior on the TUAB dataset.

3.1 Preliminary Analyses and Motivation for Spatial Hierarchy

To evaluate whether EEG abnormalities can be distinguished using non-spatial features alone, channel-wise temporal descriptors (Hjorth mobility, Hjorth complexity, and signal entropy) were extracted from 1,500 TUAB recordings for preliminary analysis and aggregated across channels. Substantial overlap between normal and abnormal EEGs was observed, indicating that temporal statistics alone lack sufficient discriminative power.

Each EEG was therefore represented as a functional connectivity graph, where nodes correspond to electrodes and edges encode phase-based synchrony. Mean adjacency matrices were computed for each electrographic subtype and compared against the normal baseline using Δ -connectivity maps:

$$\Delta M_{\text{subtype}} = \bar{A}_{\text{subtype}} - \bar{A}_{\text{normal}} \quad (1)$$

As shown in Figure 3, distinct subtype-specific spatial deviations emerge, including localized fronto-temporal hyperconnectivity, lateralized weakening in focal abnormalities, and global reductions in long-range connectivity for diffuse slowing. These observations demonstrate that clinically meaningful EEG abnormalities are primarily encoded in spatial interaction patterns, motivating hierarchical graph-based modeling.

3.2 Model Performance and Optimization

The full gated graph transformer achieved an accuracy of 0.7152, with macro-averaged precision, recall, and F1 score of 0.72, 0.71, and 0.71, respectively. Performance was higher for normal EEGs (recall 0.78) than abnormal EEGs (recall 0.64), revealing a 14-point recall gap consistent with the heterogeneous and transient nature of abnormal EEG patterns.

Algorithm 1 Spatial Multi-Expert Graph Transformer for Dynamic EEG Abnormality Detection

Require: EEG recording X , number of experts K , number of temporal segments S

Ensure: Global abnormality label $\hat{y}$ and subtype probabilities p_{sub}

- 1: Preprocess EEG signal and obtain matrix X
 - 2: **Dynamic Graph Construction**
 - 3: Split X into S temporal segments
 - 4: **for** each segment W_t **do**
 - 5: Compute functional connectivity A_t using wPLI
 - 6: Compute channel features $F_t \leftarrow [\text{mean}(W_t), \text{std}(W_t), \text{ptp}(W_t)]$
 - 7: Construct graph frame $G_t = (\mathcal{V}, A_t, F_t)$
 - 8: **end for**
 - 9: **Hierarchical Graph Encoding**
 - 10: **for** each graph G_t **do**
 - 11: Encode graph to embedding h_t
 - 12: **end for**
 - 13: **Temporal Modeling**
 - 14: Apply TransformerEncoder to $\{h_t\}_{t=1}^S$ to obtain sequence representation h_{seq}
 - 15: **Multi-Expert Reasoning**
 - 16: Compute expert weights $g = \text{softmax}(W_g h_{\text{seq}})$
 - 17: **for** each expert $k = 1, \dots, K$ **do**
 - 18: Compute expert output $z_k = \text{Expert}_k(h_{\text{seq}})$
 - 19: **end for**
 - 20: Fuse experts: $z_{\text{fused}} = \sum_{k=1}^K g_k z_k$
 - 21: **Prediction**
 - 22: Compute subtype probabilities $p_{\text{sub}} = \text{softmax}(W_{\text{sub}} h_{\text{seq}})$
 - 23: $\hat{y} \leftarrow \arg \max(z_{\text{fused}})$
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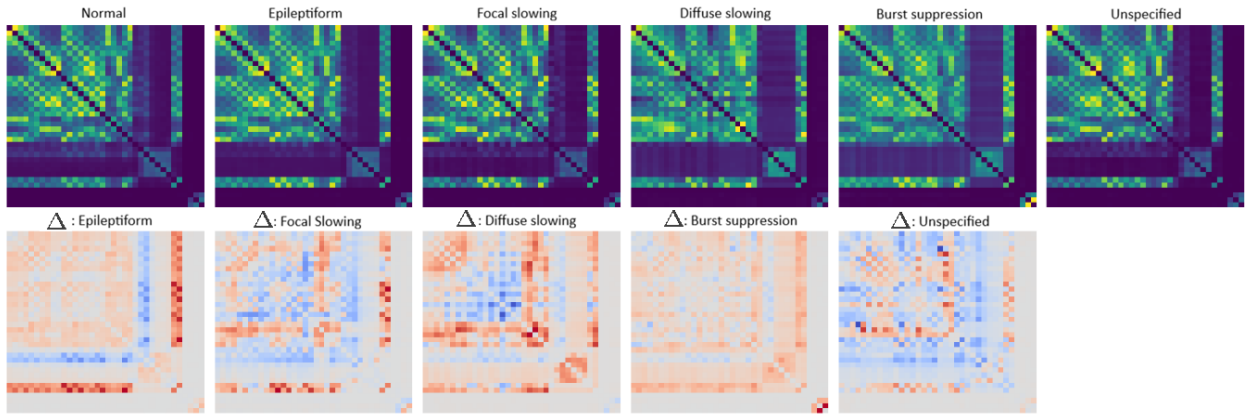


Figure 3: Mean connectivity (top) and Δ -connectivity (bottom) maps comparing normal and abnormal EEG subtypes. Distinct connectivity fingerprints particularly in focal slowing demonstrate spatially structured deviations from the normal baseline.

The reproducible ablation variants focus on feature-only modeling, hierarchical graph encoding, and adaptive expert gating. This allows the analysis to emphasize stable comparisons across the reported model components.

Ablation Analysis. Table 1 reports the reproducible ablation variants used to assess the contribution of graph-based modeling and adaptive expert gating. Feature-only modeling (A0) yields competitive accuracy but lower abnormal

Table 1: Ablation study for TUAB abnormal EEG detection using reproducible model variants.

Model Variant	Graph	Hierarchy	Gating	Acc.	N-P	N-R	A-P	A-R
A0: No Graph (Feature-Only)	No	No	Yes	0.7027	0.67	0.79	0.74	0.60
A3: Graph + Hierarchy (Single-Expert)	Yes	Yes	No	0.7027	0.71	0.70	0.69	0.71
Full Model: Graph + Hierarchy + Gating	Yes	Yes	Yes	0.7152	0.69	0.78	0.74	0.64

recall, suggesting that channel-wise temporal statistics alone may not fully capture heterogeneous abnormal patterns. Adding hierarchical graph encoding in a single-expert setting (A3) improves recall balance, indicating that spatial connectivity structure contributes to abnormal-class sensitivity. The full model combines hierarchical graph encoding with adaptive expert gating and achieves the highest overall accuracy while maintaining a balanced precision–recall profile.

3.2.1 Threshold Sensitivity and Recall–Precision Trade-offs

While the multi-expert model improves overall performance and abnormal-class precision, a modest reduction in abnormal recall is observed at the default decision threshold of 0.5 (Table 1). To determine whether this reflects a loss of discriminative capacity or a shift in operating point, we performed a post-hoc threshold sensitivity analysis without retraining.

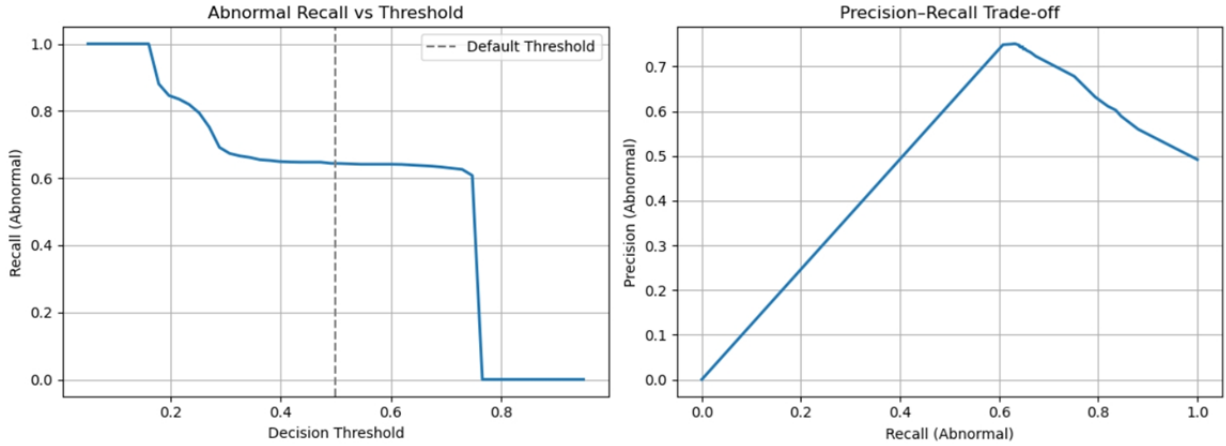


Figure 4: Post-hoc threshold analysis of the gated model. Abnormal recall versus decision threshold (left) and corresponding precision–recall trade-off (right).

Post-hoc Expert Contribution Analysis. Post-hoc analysis of the gating mechanism shows high-entropy expert utilization under weakly structured inputs, indicating stable routing rather than expert collapse. In addition, average gating distributions differed across abnormal subtypes, suggesting emergent expert specialization even without explicit subtype-wise supervision. Expert-level ablations requiring retraining are deferred to future work.

Hyperparameter tuning. Limited tuning did not yield consistent gains; default settings are used throughout.

Optimization sensitivity. We evaluated multiple optimizer, activation, and learning-rate scheduling configurations while fixing the architecture. Performance differences were modest across settings, suggesting that the model remains sensitive to optimization choices and that architectural comparisons should be interpreted with this variability in mind.

3.3 Interpretation

Overall, the results indicate that representing EEG recordings as dynamic spatial connectivity graphs enables interpretable analysis of abnormal EEG patterns, with distinct electrographic subtypes exhibiting characteristic connectivity differences. The multi-expert transformer facilitates subtype-aware reasoning and adaptive decision fusion, while threshold analysis highlights the importance of operating-point selection for balancing precision and recall. Observed recall variability reflects clinical heterogeneity and operating-point selection rather than loss of discriminative signal.

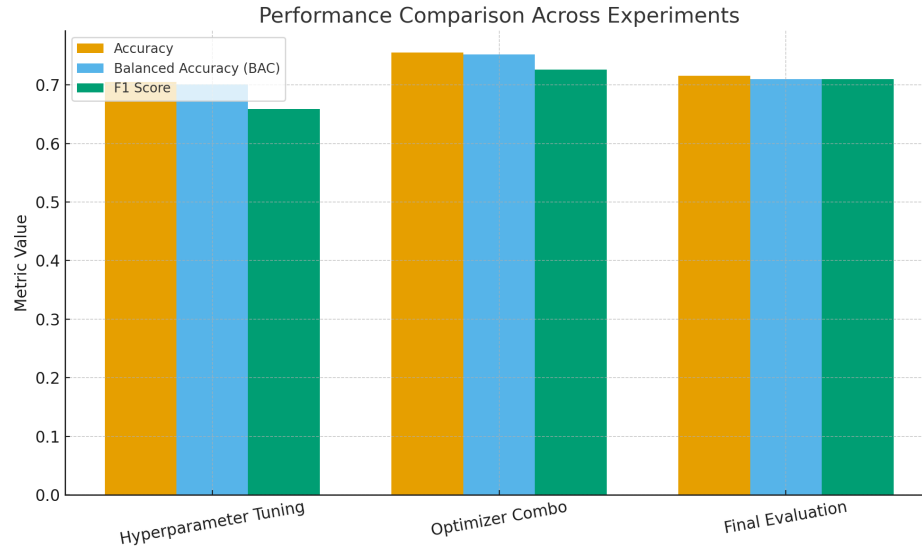


Figure 5: Performance comparison across tuning, optimization, and final evaluation stages.

4. Conclusion and Future Work

This work presents a Spatial Multi-Expert Graph Transformer for EEG abnormality detection that models EEG signals as dynamic connectivity graphs with hierarchical encoding for subtype-aware and interpretable predictions. The framework achieves competitive performance while remaining sensitive to dataset-specific preprocessing and electrode layout variability.

Future work will focus on incorporating anatomically informed hierarchical pooling to better align graph abstractions with neurophysiological structure, beyond purely data-driven pooling methods. In addition, recall-aware expert gating strategies will be explored to prioritize abnormal-class sensitivity under clinically asymmetric error costs. Extensions toward causal interpretability and multimodal integration will also be investigated.

5. Competing interests

The authors declare that there is no competing interest in this study.

References

- [1] K. Blinowska and P. Durka, "Electroencephalography (eeg)," *Wiley encyclopedia of biomedical engineering*, vol. 10, p. 9780471740360, 2006.
- [2] I. Sanchez Fernandez, T. Loddenkemper, A. S. Galanopoulou, and S. L. Moshé, "Should epileptiform discharges be treated?," *Epilepsia*, vol. 56, no. 10, pp. 1492–1504, 2015.
- [3] Z. Li, W. Zhao, J. Xu, and P. Liu, "Ffcl: Feature fusion of cnn and lstm for abnormal eeg detection," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 1745–1756, 2022.
- [4] W. Jing, K. Zhou, L. Zhang, and X. Chen, "Sparcnet: Spectral–temporal residual convolutional network for eeg classification," *Frontiers in Neuroscience*, vol. 17, p. 1208754, 2023.
- [5] J. Song, C. Wang, Z. Lu, and F. Zhang, "S3t: Spatial–spectral–temporal transformer for eeg classification," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1234–1238, IEEE, 2021.
- [6] X. Yang, H. Zhang, Q. Wang, and Z. Zhang, "Biot: Biosignal transformer for clinical eeg interpretation," *Neurocomputing*, vol. 567, p. 127018, 2024.
- [7] H. Jiang, Q. Liu, Y. Zhang, and W. He, "Labram: Large-brain transformer for eeg foundation model pretraining," *IEEE Transactions on Biomedical Engineering*, vol. 71, no. 6, pp. 1902–1915, 2024.
- [8] R. Tang, X. Zhou, H. Wang, and J. Chen, "Corr-dcrnn: Correlation-aware diffusion convolutional recurrent network for eeg representation learning," *Pattern Recognition*, vol. 133, p. 108980, 2022.
- [9] R. Ding, Y. Zhang, Q. Li, and T. He, "Lggnet: Local-global graph neural network for eeg-based abnormality detection," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 5, pp. 2101–2114, 2024.
- [10] Y. Guo, X. Zhang, C. Wang, J. Li, and P. Zhou, "Xaiguiformer: Explainable transformer with connectome tokenization for eeg disorder identification," *Neural Networks*, vol. 180, pp. 106–123, 2025.
- [11] M. Hardmeier, F. Hatz, H. Bousleiman, C. Schindler, C. J. Stam, and P. Fuhr, "Reproducibility of functional connectivity and graph measures based on the phase lag index (pli) and weighted phase lag index (wpli) derived from high resolution eeg," *PloS one*, vol. 9, no. 10, p. e108648, 2014.
- [12] I. Obeid and J. Picone, "The temple university hospital eeg data corpus," *Frontiers in Neuroscience*, vol. 10, p. 196, 2016.